Fixed-seed spacing on an interior record-atom tail

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Let

$$x_{t+1} = U_+(x_t) := \frac{3x_t + 1}{2^{a_t}}, \qquad a_t = v_2(3x_t + 1), \qquad A_t = \sum_{h<t} a_h, \tag{1}$$

be an injective infinite accelerated orbit of positive odd integers. Put

$$Q_t := \frac{3^t}{2^{A_t}}, \qquad \mathcal{K} := \lim_{t\to\infty} \frac{x_t}{Q_t} < \infty, \tag{2}$$

and suppose that $Q_t \to \infty$. Let

$$j_0 < j_1 < j_2 < \cdots \tag{3}$$

be consecutive coefficient tail records. Assume that every record gap from j_0 onward has one of the four exponent words

$$\mathcal{A}_4 := \{1, 12, 1113, 1122\}. \tag{4}$$

Write $\omega_i \in \mathcal{A}_4$ for the gap from j_i to j_{i+1}. For an atom w, let $r(w)$ be its length and $B(w)$ its total exponent; thus

w	1	12	1113	1122
$r(w)$	1	2	4	4
$B(w)$	1	3	6	6.

(5)

For a finite atom word $V = w_0 \cdots w_{L-1}$, put

$$r(V) := \sum_{h<L} r(w_h), \qquad B(V) := \sum_{h<L} B(w_h). \tag{6}$$

Theorem 1 (record-atom return lattice and spacing)**.** *Suppose that a length-L atom factor repeats at record positions $i < k$:*

$$\omega_i\omega_{i+1}\cdots\omega_{i+L-1} = \omega_k\omega_{k+1}\cdots\omega_{k+L-1} = V. \tag{7}$$

Then

$$\boxed{2^{B(V)+1} \mid x_{j_k} - x_{j_i}, \qquad 2\,3^{r(V)} \mid x_{j_{k+L}} - x_{j_{i+L}}.} \tag{8}$$

The two differences are positive. If

$$\alpha := \log_2 3, \qquad C_0 := \log_2(\mathcal{K}Q_{j_0}), \tag{9}$$

then every repetition in (7) satisfies

$$\boxed{B(V) + 1 < C_0 + (\alpha - 1)k.} \tag{10}$$

Equivalently,

$$\boxed{k > \frac{B(V) + 1 - C_0}{\alpha - 1}.} \tag{11}$$

Thus a repeated atom factor of dyadic weight $B(V)$ cannot occur before the fixed ordinary ray has accumulated comparable record-index surplus.

Proof. Let v be the raw exponent word obtained by concatenating the atoms in V, and put $r = r(V)$ and $B = B(V)$. There is an integer carry $C(v)$ such that each occurrence obeys

$$2^B x_{j_i+r} = 3^r x_{j_i} + C(v), \qquad 2^B x_{j_k+r} = 3^r x_{j_k} + C(v). \tag{12}$$

The terminal states in (12) are odd. Reduction modulo 2^{B+1} gives

$$3^r x_{j_i} + C(v) \equiv 2^B \equiv 3^r x_{j_k} + C(v) \pmod{2^{B+1}}. \tag{13}$$

Since 3 is a unit modulo 2^{B+1}, the first divisibility in (8) follows. Subtracting the two identities in (12) gives

$$2^B\big(x_{j_{k+L}} - x_{j_{i+L}}\big) = 3^r\big(x_{j_k} - x_{j_i}\big). \tag{14}$$

The first divisibility in (8), substituted into (14), proves the second.

The normalized height x_t/Q_t is strictly increasing, because

$$\frac{x_{t+1}}{Q_{t+1}} - \frac{x_t}{Q_t} = \frac{1}{3Q_t} > 0. \tag{15}$$

Also $Q_{j_k} > Q_{j_i}$ for $k > i$. Hence $x_{j_k} > x_{j_i}$, and (14) gives positivity at the endpoints as well. Therefore

$$2^{B+1} \le x_{j_k} - x_{j_i} < x_{j_k} < \mathcal{K} Q_{j_k}. \tag{16}$$

For consecutive coefficient records one has

$$1 < \frac{Q_{j_{h+1}}}{Q_{j_h}} \le \frac{3}{2}. \tag{17}$$

Iteration of (17) yields

$$Q_{j_k} < Q_{j_0} \left(\frac{3}{2}\right)^k. \tag{18}$$

Substituting (18) into (16) and taking binary logarithms proves (10), and (11) is its rearrangement. □

Let $p_\omega(L)$ denote the number of distinct length-L factors of the infinite atom word $\omega_0\omega_1\omega_2\cdots$.

Corollary 2 (stationary atom-complexity floor)**.** *For every $L \ge 1$,*

$$\boxed{p_\omega(L) > \frac{L + 1 - C_0}{\alpha - 1}.} \tag{19}$$

In particular,

$$\boxed{\liminf_{L\to\infty} \frac{p_\omega(L)}{L} \ge \frac{1}{\log_2 3 - 1} = 1.709511\ldots.} \tag{20}$$

Consequently the record-atom word is neither eventually periodic nor quasi-Sturmian.

Proof. Among the first $p_\omega(L)+1$ factors of length L, two are equal. For the second occurrence in such a pair, $k \le p_\omega(L)$. Every atom has total exponent at least one, so its common factor V satisfies $B(V) \ge L$. Equations (10) and (7) therefore give

$$L + 1 < C_0 + (\alpha - 1)p_\omega(L),$$

which is (19). Division by L and passage to the lower limit proves (20). An eventually periodic word has bounded factor complexity, while a quasi-Sturmian word has $p_\omega(L) = L + O(1)$; both contradict (20). □

Corollary 3 (bounded one-atom runs)**.** *Suppose that every run of the one-letter atom* 1 *has length at most* M*. Then*

$$\boxed{p_\omega(L) > \frac{L + 2\lfloor L/(M+1)\rfloor + 1 - C_0}{\alpha - 1}.} \tag{21}$$

Consequently,

$$\boxed{\liminf_{L\to\infty} \frac{p_\omega(L)}{L} \geq \frac{1 + 2/(M+1)}{\log_2 3 - 1}.} \tag{22}$$

Proof. Any length-L factor contains at least $\lfloor L/(M+1)\rfloor$ atoms different from 1. Replacing a one-letter atom by any other atom raises its total exponent from 1 to at least 3. Hence every such factor satisfies

$$B(V) \geq L + 2\left\lfloor \frac{L}{M+1} \right\rfloor. \tag{23}$$

Apply the proof of Corollary 2 with (23) in place of $B(V) \geq L$. This proves (21), and (22) follows after division by L. □

Corollary 4 (sharpened interior-tail obstruction)**.** *Every normalized hypothetical divergent orbit has at least one of the following properties.*

1. *There are infinitely many deficit-four horizon exits.*
2. *From some coefficient record onward, all gaps are in* $\mathcal{A}_4$*, the synchronized companion is bounded by*

$$1 < e_t \leq \frac{41}{3}, \tag{24}$$

and its ceiling defects satisfy

$$d_t \in \{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7\}. \tag{25}$$

Its logarithmic clock grows at least $i\log(9/8) - O(1)$ *at the* i*th record, and its atom word obeys (19)–(20). If the runs of the atom* 1 *are bounded by* M*, then it obeys the stronger estimates (21)–(22).*

Proof. The deficit-four record-renewal dichotomy gives the first alternative or the four-atom tail in the second. The synchronized-companion confinement theorem gives (24), the finite defect alphabet, and the logarithmic-clock bound on that tail. Corollaries 2 and 3 supply the fixed-seed spacing and complexity conclusions. □